



OPEN Technical and economic simulation of a hybrid renewable energy power system design for industrial application

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Countries in Africa rank the highest in low energy accessibility and air pollution globally. Presenting the urgent need to explore renewable energy sources to tackle the power challenge and reduce the carbon footprint for a greener atmosphere. A novel hybrid wind and solar renewable energy power system (HREPS) coupled to a battery that is capable of powering industrial appliances in the Basse district of The Gambia has been proposed. The HREPS size was estimated automatically with the PVsyst software taking into account the PV power and battery capacity based on the load profile and loss of load probability, with all the necessary meteorological, technical, and economic factors. The optimal size was possible with 20 photovoltaic modules (250 W) and 1 Wind generator (1 KW) compatible with the annual load requirement of about 2555 MWh. These results suggested that the HREPS is reliable for the user's load demand in all the tested meteorological conditions of the study area. When the HREPS is linked to any other renewable source, it shows the potential of gaining about 8395 kWh annually in the battery. The continuous operation of the hybrid system for 21 years presents a net gain of > 400% for the standalone systems with a projected gain of > 600% when linked to a small smart grid system. Since high taxes may scuttle these gains, it is recommended that the Government encourage investment in renewable energy power systems by subsidizing the taxes on the purchase of items to make them more affordable.

Keywords Solar-wind hybrid, PVsyst simulation, Cost implication analysis, Industrial electrification, Basse Santa Su District

Abbreviations

GlobHor	Global horizontal irradiation
GlobEff	Effective Global, corr. for IAM and shadings Available Solar Energy
DiffHor	The horizontal diffuse irradiation
DiffEff	Effective diffuse irradiation
DiffInc	Incident diffuse irradiation
E_Avail	Available Solar Energy
EUnused	Unused energy (battery full)
E_Miss	Missing energy
E_User	Energy supplied to the user
E_Load	Energy need of the user (Load)
SolFrac	Solar fraction (EUsed / ELoad)
HREPS	Hybrid wind and solar renewable energy power system
MWh	Megawatt hour
kWh	Kilowatt hour
MPPT	Maximum Power Point Tracking
$P_{WT}(t)$	Wind turbine power
$P_{WT}^{av}(t)$	Average wind power generated by the turbine
c_p	The Betz coefficient

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P_{pv}	Solar panel peak power output
I_T	The radiation on the solar panel in units of kWh/m ²
I_S	The reference value of optimum irradiation (1000 W/m ²)
Y_{pv}	The panel nominal power
T_{amb}	Stable ambient temperature
T_C	Solar cell temperature in Kelvin
G_S	The site-specific solar irradiation
NO_{CT}	The characterized solar cell temperature

Accessing modern electricity that is reliable, cheaper, and sustainable remains a dream far from being realized for about 620 million of the world's population in the next decade. In Sub-Saharan Africa (SSA), the International Energy Agency¹ reported that more than 50% of the inhabitants had no access to clean energy. In addition, they predicted that electricity demand will rise by approximately 4.6% by the beginning of 2022 with the support of industrial and economic activities². Driven by the quest for industrialization, and rising population, SSA needs to seek power alternatives to better its electricity-generating capacity. Furthermore, fears of the impacts of overdependence on fossil fuels by the remote and urban dwellers, as well as the industrial sector on the environment require urgent attention³.

A hybrid renewable energy power system comprising wind and solar energy can compensate for the energy deficit of many nations and it remains a pragmatic step to exploit these renewable sources and ensure their integration for a grid-connected and or stand-alone application⁴. Hybridization of renewable sources of power with the actual battery capacity is vital in addressing the problem of intermittency of power output and overdependence on the nation's power grid⁵.

Wind and solar power systems are accepted universal renewable options for producing electricity and tackling climate change. These sources are readily available at no cost and have low environmental threats. In power generation for rustic and industrial applications, their integration remains advantageous both technically and economically^{6,7}.

In practice, hybrid system under-sizing usually affects the smooth operation of all the industrial-electric appliances, and over-sizing the hybrid system may become an economic liability. Therefore, technical and economic simulation of the hybrid renewable power system can assist in cutting costs and guarantee power solidity⁸.

Most of the industrial operations available in developed and developing countries use a high proportion of energy that emits large concentrations of gaseous and particulate pollutants that are linked to global warming⁹. The need to address this challenge has led to the 2015 Paris Conference attended by the countries of the five continents (Africa, South America, Europe, North America, and the Middle East). At the conference, it was agreed that measures to tackle climate change and limit the world temperature increase to 2 was adopted to tackle the climate change effect¹⁰. To reduce the impact of climate change, they made deliberations and suggestions on how to invest in the renewable energy sector and convert the pollution systems into those with low carbon emissions and environmental nuisance. To achieve this¹¹, reported that renewable energies can conveniently replace high-carbon emission fuels and boost a country's economy through electricity sales. Solar, wind, geothermal, and biomass are well-known renewables in addition to the old scientific hydraulic applications.

The transition from these fossil fuels to clean energy depends on the renewable energy generation and technologies which are active drivers of the transition⁶. Over the past two decades, green energies were coated to have a high cost of production making it impossible to be connected to a network or have a single appliance, and their use for commercial purposes was also limited¹². Recently, the prices of most of the equipment for renewables have been reduced in many nations given the opportunities to afford new ones for renewable energy production¹³.

Hybridization or integration of renewable energy sources and power storage is a bold step toward achieving a reliable, affordable, and well-planned renewable energy power system^{14,15}. Furthermore, accessing these technologies at affordable prices, and the confidence of having the needed skills to carry out the installations has been boosted recently, linking solar photovoltaic and wind power increase rapidly worldwide¹⁰.

Few studies have been conducted on optimizing the design of renewables grid-connected or stand-alone for residential use in Sub-Saharan Africa. Some of the studies are as follows:

Samy¹⁶ conducted a techno-economic assessment of hybrid renewable energy systems in the Monshaet Taher area of Egypt. He highlighted the potential of applying the renewable energy-hydrogen concept for the electrification of villages by studying five different hybrid power system configurations using the HOMERTM software.

Mulumba and Farzaneh¹⁴ carried out a dynamic techno-economic simulation and analysis of a hybrid renewable energy power system in the remote areas of Kenya using a multiple objective optimization modeling technique and the epsilon (E) approach. They reported that the optimal size of the hybrid renewable energy system was feasible at 330 W for 26 photovoltaic panels and 3 (1kw) wind turbines sufficient for 37.94 MWh annual loads.

Eze et al.¹⁷ assessed the technical and economic feasibility of a hybrid renewable energy system in the Institutional Buildings of Kenya. They calculated the 25-year current cost to determine the lowest energy cost for the Diesel/Photovoltaic/Grid sources to realize the optimized design system.

In addition, several investigations adopting different technologies nowadays have been published from which the following attract more attention:

Sharma et al.¹⁸ compare simulation software for hybrid energy systems with evaluation approaches. The creation of specialist tools like Homer Pro, Energy Pro, iHOGA, and TRNSYS to study these systems stands

out in the previous decade. The study thoroughly analyzes these various instruments in terms of economic and environmental assessment and found that only one considers techno-economic effects.

Similarly¹⁹, used a methodology that tackles the variability issues with many renewable energy sources by using one as the foundation for the less variable ones. To boost stability and ensure the continuation of electrical service, they recommend developing a hybrid system that incorporates hydroelectric power or hydroelectric energy storage. All these studies recommended software simulation for conducting techno-economic analyses of hybrid renewable energy power systems worldwide. However, most of these studies emphasized quantitative data without much regard for meteorological and climatic conditions affecting renewable energy power systems.

Ayua et al.²⁰ reported that The Gambia has a surplus of agricultural products such as fruit juice, confectionery nuts, chicken feeds, and fish which are inadequately utilized because of the unavailability of the desired processing industries. Investment in the industries that process these products will reduce the nation's overdependence on imports and increase its exports.

The present study presented a simulated hybrid renewable energy power system combining wind and solar sources to power solar water heaters for heating and processing food products, solar cooking and cooling systems for mitigating the effects of greenhouse gases, and can also be applied for solar-powered cooling systems for space cooling. The hybrid system design considers the climatic variables' effects to dampen the meteorological conditions and climatic factors that cause power intermittencies to enhance the effectiveness and reliability of renewable energy systems for agriculture and other industrial applications. The innovative renewable hybrid energy power system proposed is intended to show an optimum hybrid structure that guarantees a steady power supply while lowering the amount of nonrenewable energy power options. The system reduces the financial load on the Basse Santa Su community by providing significant cost reductions. Investor risk may be decreased, immediate dependability and affordability can be guaranteed, and profit maximization can be achieved with the proposed novel hybrid technology. It is hoped that adopting it will improve the poor air quality of the site locations to safer levels. Finally, the available primary renewable energy resources converted to electrical energy, and contrasted with the environment can boost industrial production and create job opportunities.

Materials and methods
Description of the methodology and study site

The study employed the solar-wind hybrid power system generation for industrial application in the Basse Santa Su district of The Gambia using the PVsyst tool and wind generator model based on the available meteorological dataset of the location. The PVsyst is widely applied due to its robustness. It can allow simulations under different conditions and remains a good option for hybrid standalone or grid-integrated power system designs. The software is capable of carrying out the needed economic analysis of the energy production to power industrial machines like boilers and or water heaters proposed in this study. The wind speed data generated by the PVsyst for the location was used to calculate the wind turbine power output using the Betz model as in Eqs. 1–3. Finally, the sizing and economic simulation of the proposed system were automatically computed with the PVsyst based on the input data in Table 1. Figure 1 presents a schematic representation of the proposed hybrid renewable power system and the study site.

The Gambia is a country that depends on agriculture and tourism as the major source of income and the Basse Santa Su area is one of the major areas for tourism and agriculture activities. However, the lack of steady and affordable electricity posed limitations to these activities. As can be seen in Fig. 1, the proposed hybrid renewable power system comprises of solar PV module, wind generator, and any other desired and available source that may be incorporated depending on the available renewable resources in the Basse area of The Gambia. Extraction of the available renewable energy sources for use in the proposed hybrid system to power

S/N	Parameters	Input/values
1	Project name	Industrial 5 kWp stand-alone
2	Site/location	Basse Santa Su-The Gambia (13.31 N, 14.22 W)
3	Field type	Fixed tilted plane
4	Field Parameters	Plane tilt 15 Azimuth 0.0(facing south)
5	System type	Hybrid Stand-alone
6	Simulation data	PVGIS TMY 5.1 meteo data
7	PV modules	Mono Crystalline
8	Number of modules	20 unit
9	Unit power	250 Wp
10	Nominal power	5.5 kWp
11	Mpp current	22.0 A
12	Mpp voltage	500 V
13	Inverter	POWMr-5000
14	Number of inverters	1 unit
15	Inverter unit power	5.0 kW
16	Pnom AC of inverter	5.0 kW

Table 1. PVsyst Simulation parameters for the proposed power system.

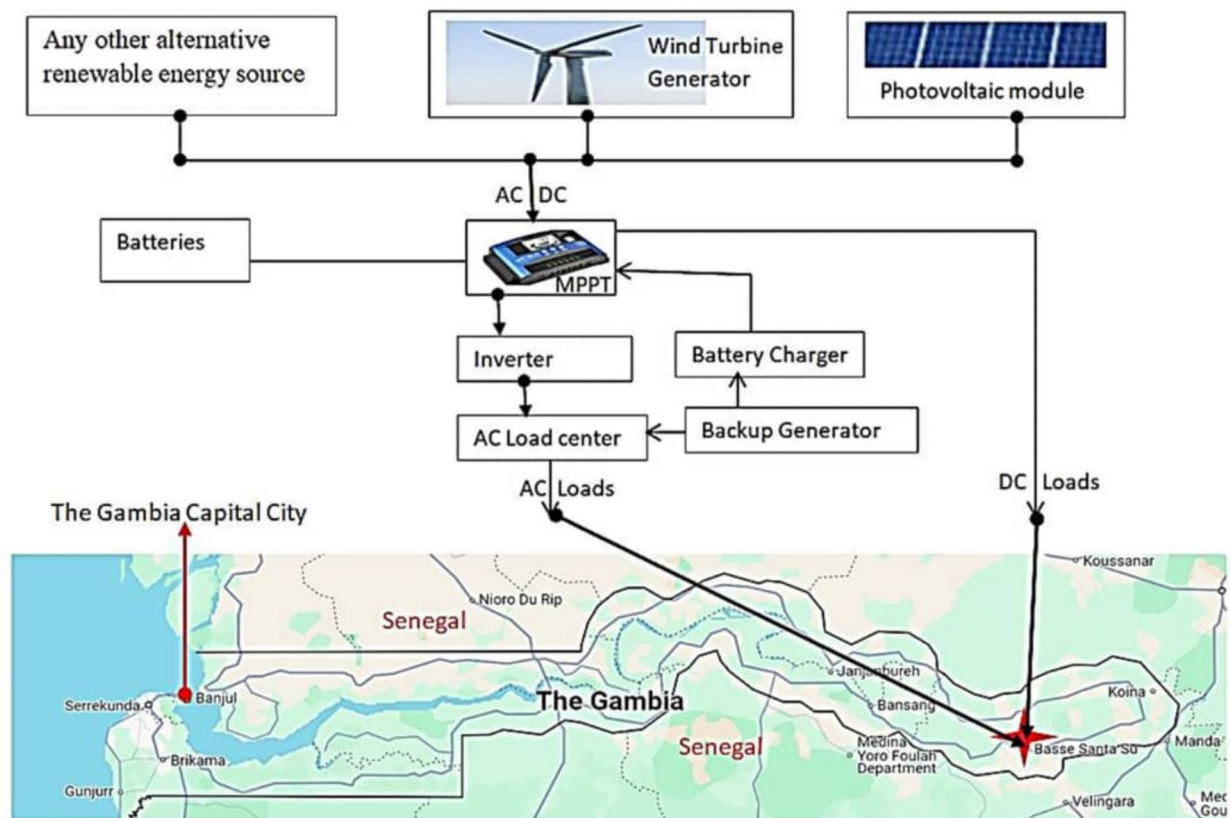


Fig. 1. Schematic representation of the proposed standalone hybrid renewable energy power system and map of The Gambia showing the location of Basse Santa Su.

industrial water heaters and or boilers in Basse for agricultural products processing, and green energy sources for a better atmosphere is the novelty of our work.

Design and optimization procedures

For the stand-alone system proposed in this study, the PVsyst tool allows automatic sizing of the PV power and battery capacity based on the load profile and loss of load probability. The software easily simulates the optimal values for the needed design parameters to be installed. It automatically runs a series of simulations in which one or more parameters are methodically changed within a predetermined range. Every simulation's output is saved and accessible through a different tab that plots various simulation variables as a function of the changed parameters. The renewable sources of the proposed hybrid power system and materials are highlighted below.

Wind generator

The wind turbine can generate power depending on the wind speed data, the Betz coefficient, and the density of air in an area, and the power P_{WT} can be mathematically expressed as in Eq. 1.

$$P_{WT}(t) = \frac{1}{2} c_p(\alpha, \gamma) \rho_a A V^3 \quad (1)$$

where c_p is the Betz coefficient depending on the function angles α, γ , ρ_a is the air density, A is the area of the turbine in units of m^2 and V is the input wind speed^{10,21}.

The turbine function angle α in Eq. (1) is defined according to Eq. 2.

$$\alpha = \frac{R \omega_m}{V} \quad (2)$$

where R denotes the radius of the turbine and ω_m is the turbine shaft velocity angle. In practice, the wind power delivered by a turbine generator depends on the wind speed which is usually limited within some range as presented in Eq. 3.

$$P_{WT}^{av}(t) = \frac{1}{2} c_p(\alpha, \gamma) \rho_a A V^3(t) \text{ if } v_j \leq V \leq V_r \quad (3)$$

where V_r is the turbine-rated wind speed of operation, and V_j is the minimum required wind speed.

The interpretation of the average power generated by the turbine in Eq. 3 is that, for $V < V_r$ and $V > V_c$, $P_{WT}^{av}(t) = 0$. While $P_{WT}^{av}(t) = P_W^i$ if $V_i < V < V_c$. V_c is the cutting or stopping wind speed of the turbine for safety.

Photovoltaic generator

The industrial appliances such as water heaters and or boilers to be powered by the proposed hybrid system in Basse consider solar energy as a reference point within the specification, given the Basse site characteristics that meet the conditions of harvesting the solar energy sustainably for the application. Investing in solar photovoltaic is based on the peak power output P_{pv} from the solar panels which is given by Eq. 4.

$$P_{pv} = \frac{I_T}{I_S} [Y_{pv} \times f_{pv}] \quad (4)$$

where I_T denotes the radiation on the solar panel in units of kWh/m², I_S is the reference value assumed to be the optimum irradiation and is usually taken as 1000 W/m², Y_{pv} is panel nominal power and f_{pv} stands for the loss parameters such as aerosol effects on the solar panel surface, variations in ambient temperature and distance from the conductors of electricity as a result of joints between the conductors²¹.

The maximum power output is usually better at stable ambient temperature T_{amb} and it can be shown with Eq. 5 that the operational part depends on solar radiation which can contribute as follows:

$$T_c = G_s \left(\frac{NO_{CT} - 25}{1000} \right) + T_{amb} \quad (5)$$

where T_c is the temperature of the solar cell in Kelvin, G_s is the site-specific solar irradiation, and NO_{CT} represents the characterized solar cell temperature^{10,14}.

Batteries

Batteries are necessary for the proposed hybrid power system because they can store the generated energy of the system besides maintaining the needed balance between the energy produced by the hybrid system and demand. The Trojan IND29-4 V Deep-cycle batteries, which are extensively cycled at a partial state of charge and are utilized in off-grid and unstable grid applications, served as the basis for our simulation. Regularly using a battery at the partial stage of charge can drastically shorten its lifespan, necessitating expensive and frequent battery replacements. The battery has a maximum capacity of 2166 Ah, an 81% round-trip efficiency, and a minimum state of charge of 20%. The batteries have a 20-year lifespan and a 10,900 kWh lifetime throughput^{16,22}. Battery counts range from 0 to 250 which is 25 steps. The selected specification was carefully selected based on the energy demand and the battery capacity Q relationship as in Eq. 6.

$$Q = \frac{E_d A}{VT \xi_{inv} \xi_{cable}} \quad (6)$$

where E_d is the energy demand, A is the autonomy of the battery which is the number of days it can work alone without external input, V is the working voltage, T is the discharging depth of the battery, and ξ_{inv} , ξ_{cable} are the

efficiencies of the inverter and cable respectively²¹.

Inverters

As can be seen in Fig. 1, inverters are deployed to convert the center direct current DC source into alternating current AC with a high level of efficiency based on the construction conditions. The universal controller with an MPPT converter that is adaptable for all systems was deployed for the proposed hybrid system simulation. This type of inverter efficiency is automatically set according to the hybrid system specification and can be adjusted automatically if the specification changes. Its efficiency ξ_{inv} is a ratio of the output power to the input power. Mathematically the inverter efficiency ξ_{inv} is given by Eq. 7.

$$\xi_{inv} = \frac{P_o}{P_i} \quad (7)$$

where P_o and P_i are the output and input power respectively. The P_i in our case is the total power generated from the wind, solar, and any other alternative source that may be added to the proposed system.

Techno-economic analysis

The PVsyst tool was used to evaluate the simulation results of the proposed hybrid renewable energy system to identify the best technical and economical resolution. Several runs were carried out to compare the performances with other technical options for Photovoltaic systems by importing the real-time data of the cost of installing Photovoltaic modules and wind turbine into the software. The data variables of the site and system parameters for the simulation are presented in Tables 1 and 2.

Tables 1 and 2 present the general simulation parameters for Basse Santa Su which is located in The Gambia, 4 m above sea level. Its coordinates are -13.31° N and 14.22° W. The PVsyst simulation produced the weather data, which was the daily average of wind speed, temperature, and solar radiation. For our investigation, we

S/N	Parameters	Input/values
1	Power rated	1000 W
2	Startup wind velocity	1.5 m/s
3	Rated voltage	110 v
4	Maximum wind speed	50 m/s
5	Tip speed	31 m/s
6	Rated wind speed	10 m/s
7	Rotor blades radius	1.0 m
8	Fan blade number	3
9	Rated rotation speed	300 U/min

Table 2. Wind generator parameters for the proposed power system.

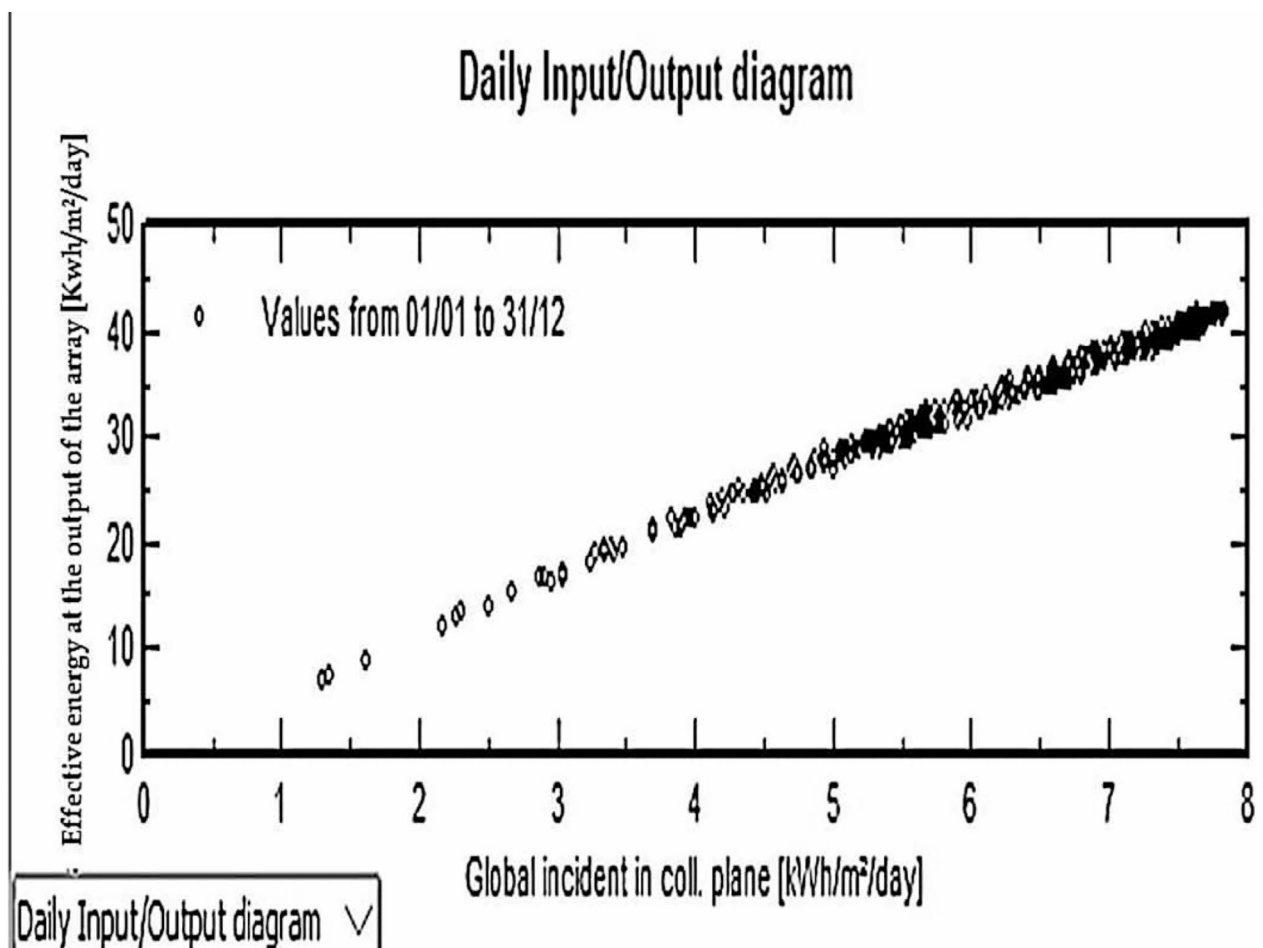


Fig. 2. Solar energy generation daily energy input and output.

selected a value of 0.2 for the Albedo effect, which typically ranges from 0.14 to 0.22 for urban locations²³. Table 2 presents the wind turbine parameters selected for the proposed hybrid renewable energy power systems. The turbine characteristics were carefully selected to meet the energy deficit of the user because the proposed solar characteristics are slightly undersized. The simulation work utilizes the feature tools from the PVsyst project design for the stand-alone system. All other known and modifiable factors, including PV module type, orientation, modular size, array structure, inverter size, and so on, were simulated for optimization purposes, while unknown parameters were left at their default settings.

Results and discussion

The effective energy realizable and the solar potential are presented in the scattered diagram in Fig. 2. It was shown that the effective energy from solar energy increases by the day. More so, with an average of 7 MW of effective energy achievable per day, solar energy can be scaled into a commercial scale. The solar fraction remains

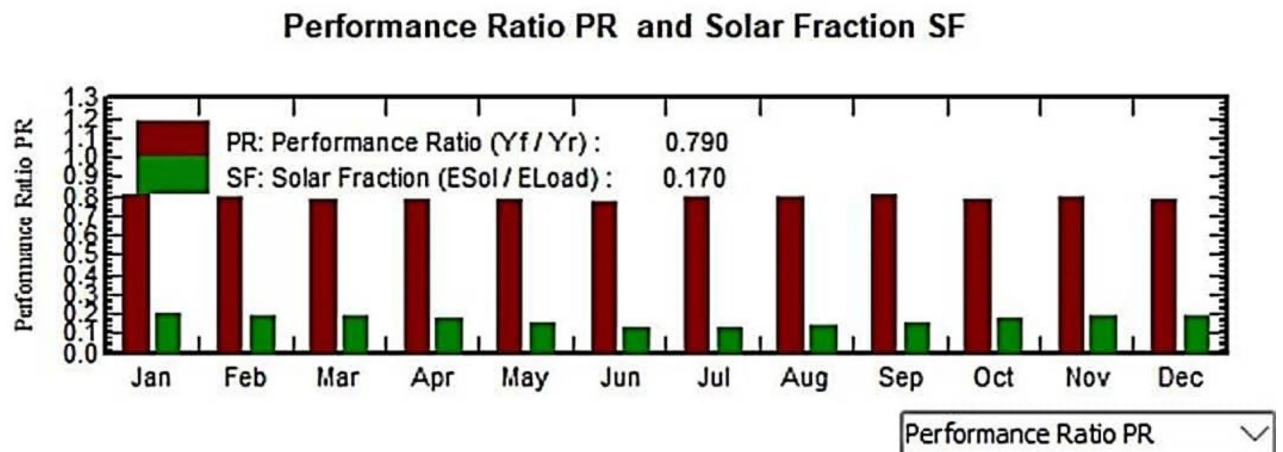


Fig. 3. Solar energy generation daily performance ratio and solar fraction.

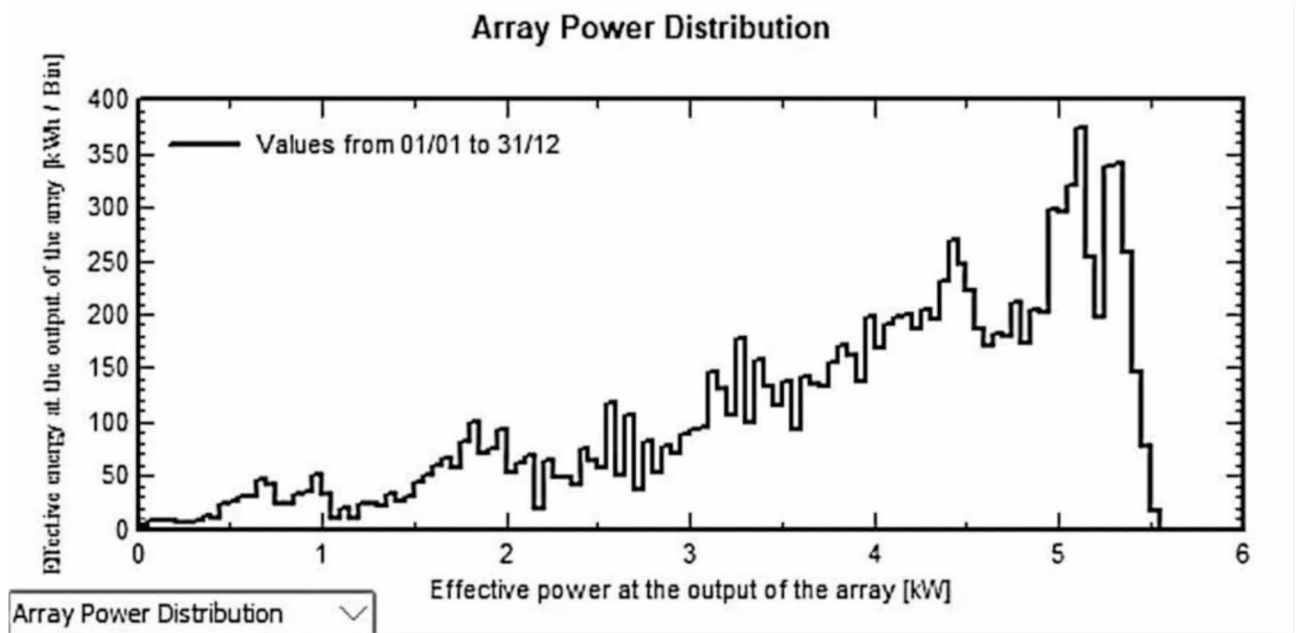


Fig. 4. Solar energy array power distribution.

constant throughout the year while the performance ratio depends on seasonal variables. A high-performance ratio is maximum in July with a build-up from April. The performance ratio converges in September through a significant constant that spans from October to March as presented in Fig. 3. The effective energy was seen to increase with respect to the effective power. The variable array power distribution increased sinusoidally to 900 kW before it dropped sharply as in Fig. 4. The implication is that the system setting will diminish due to ohmic and thermal losses which are expected to be significant. The temperature is expected to increase as the global radiation increases within the specified period as observed in Fig. 5.

The wind speed profile of the study area is presented in Fig. 6 while the power output expected from the turbine is shown in Fig. 7. It can be observed that the highest wind speed was in June and the lowest wind speed was in October with an overall mean annual wind speed of about > 3 m/s prevailing in the study area. The variation in wind speed may be attributed to the motion of the earth and atmospheric pressure differences since wind speed is proportional to the pressure gradient. The wind velocity and the terrain or topography knowledge for any site are crucial when choosing to install and operate wind generators²¹. Understanding how the wind speed influences the power output accounts for the need for the accuracy of wind velocity data in Fig. 6. Since the power generated by the moving mass of air is proportional to the rate of change of kinetic energy with time, any error associated with the wind velocity determination can affect the power output generated by the proposed HREPS significantly. Another vital tool in analyzing the efficiency of the wind turbine in the study area is the

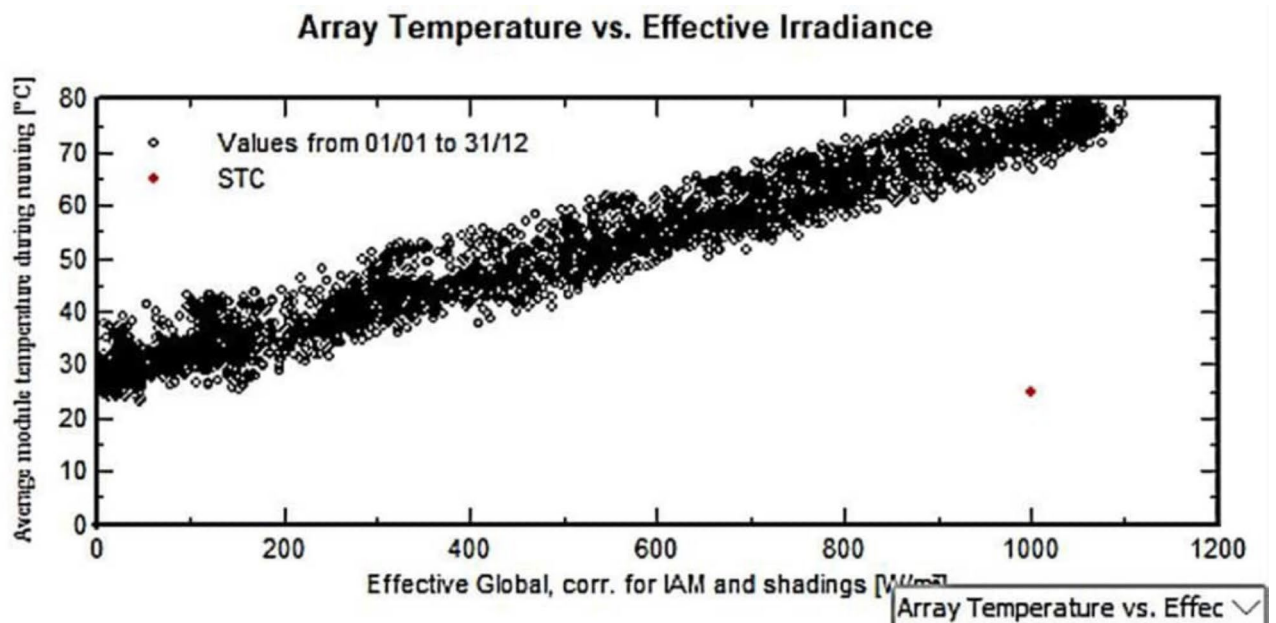


Fig. 5. Solar energy array temperature vs. effective irradiance.

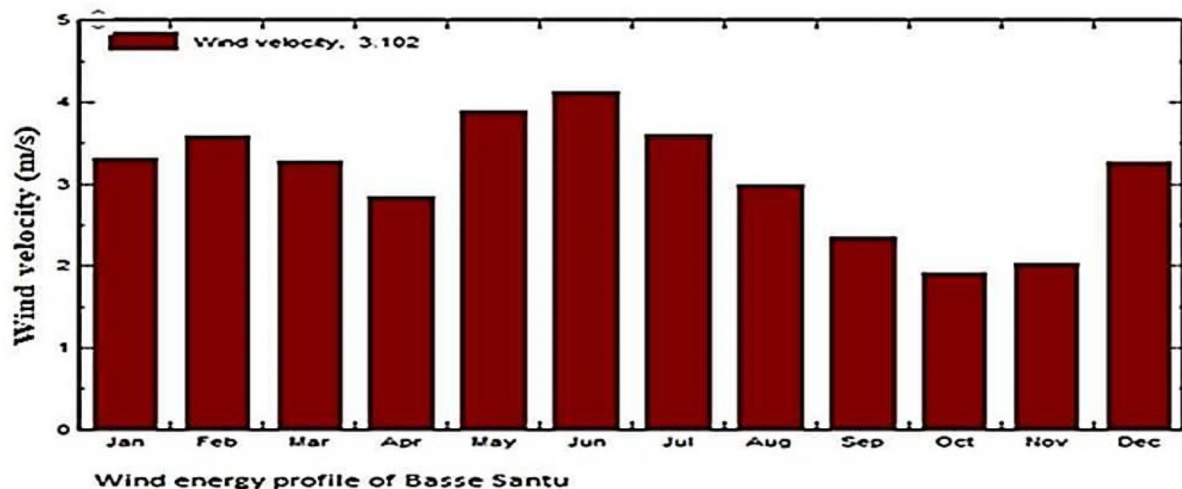


Fig. 6. Wind speed profile.

wind power curve (Fig. 7), which accounts for the wind power in the vertical and horizontal axis of the turbine blades. It can be observed that the most probable power output from the wind at the study location was recorded at wind speed of about 3–4 m/s with peak power anticipated at a wind speed of 7 m/s which was never observed in reality.

A closer look at Fig. 7 shows that the wind cut-in-speed is about > 2 m/s; at lower speeds of 0–2 m/s, the torque exerted by the wind is insufficient to rotate the turbine blades. While the rated output wind speed is observed at > 3.5 m/s. It can be seen that as the speed of the wind increases higher than the cut-in speed, the electrical power output rises sharply. However, typically somewhere > 7 m/s, the power output reaches the limit of the wind generator which is called the rated power output, and is possible only at the rated output wind speed which is not usually realized in real practice. At wind speed ≥ 8 m/s, the turbine has posed a limitation to the power output preventing further rise as a safety measure to prevent the rotor from damage. The wind analysis has further increased our confidence that the proposed HREPS can be used to achieve the energy needs of the user even when adopted and applied on a standalone base. Thus access to electricity is guaranteed as a critical component in reducing poverty in the Basse Santa Su area of The Gambia where the contribution of human advancement is typically lowest. By shifting funds from energy costs to energy productivity initiatives, job creation will be boosted and consequently, the economic life of the locals improve.

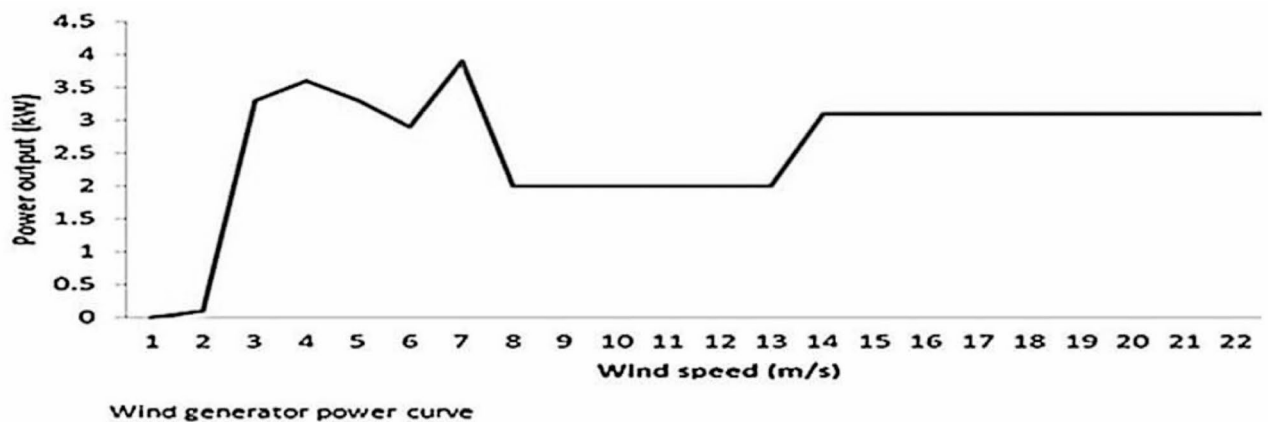


Fig. 7. Wind generator power curve at Basse Santu Su.

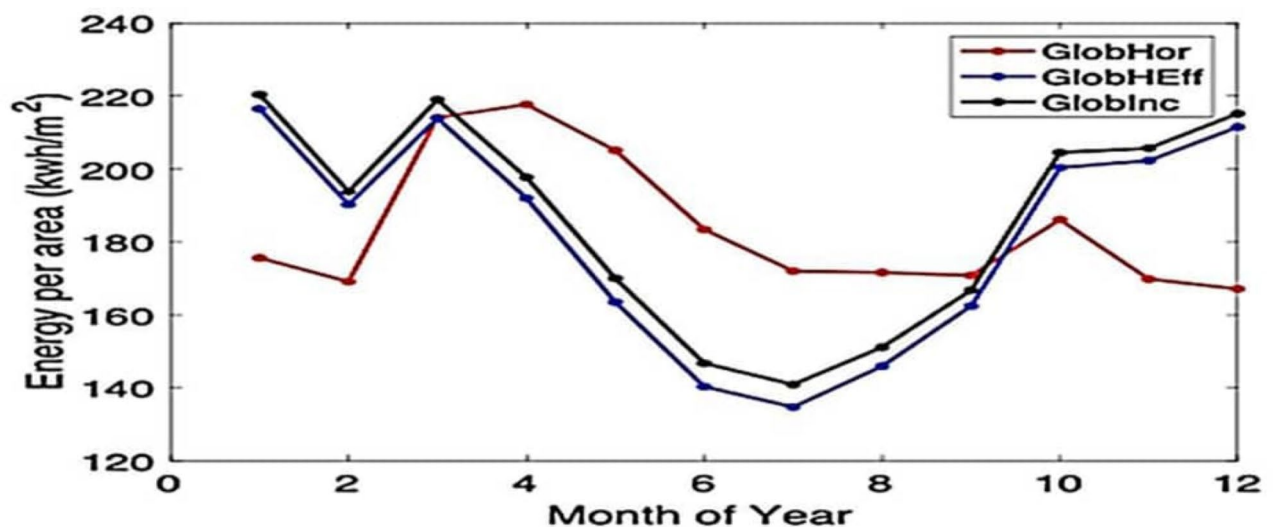


Fig. 8. Profiling of global energy generation potential of solar and wind resources.

Like the global incident energy presented in Fig. 2, the effective global (correction for IAM and shadings) tends to increase to a maximum of 220 kWh/m² as presented in Fig. 8. Secondly, it is dependent on seasonal influences i.e., it is high in the early dry season (September-October-November (SON)) and late dry season (December-January-February (DJF)). The effective global (correction for IAM and shadings) is low in the early wet season (March-April-May (MAM)) and late wet season (June-July-August (JJA)). Like the effective global (correction for IAM and shadings), the global incident in the collectors' plane is dependent on the season. The differential between the effective global and global incident is <3%. However, global horizontal irradiation (GlobInc) is not dependent on seasonal influences. In the case of the study, it is highest in April and lowest in February. Since GlobInc is the amount of terrestrial irradiance falling on a surface horizontal to the surface of the earth. Its pattern over the study area simply means that the influence of cloud formation and shading has a significant effect on the solar energy potential of the study area.

The comparative analysis of the available solar energy (E-Avail) and energy needs of the user (E-Load) is presented in Fig. 9. The energy need of the user is found to be far lower than the available solar energy. Like the effective global and global incident, the available solar energy is dependent on seasonal influences. The energy need of the user is less than 40% of the available solar energy. Hence, the rate of energy usage is presented in Fig. 9. It is shown that users in the research location use more energy in the wet season than in the dry season.

The horizontal diffuse irradiation (DiffHor), effective diffuse irradiation (DiffEff), and incident diffuse irradiation (DiffInc) are presented in Fig. 10. It is observed that the diffuse parameters are dependent on seasons i.e. they are high during raining season and low during the dry season. The value of the horizontal diffuse irradiation is almost twice the value of the effective diffuse irradiation and incident diffuse irradiation. However, the addition of effective diffuse irradiation and incident diffuse irradiation will be slightly greater than the horizontal diffuse irradiation. Hence, the vegetative cover over the research site is significant to absorb irradiation. This scenario is good for photovoltaic modules as it will certainly reduce the infrared radiations that

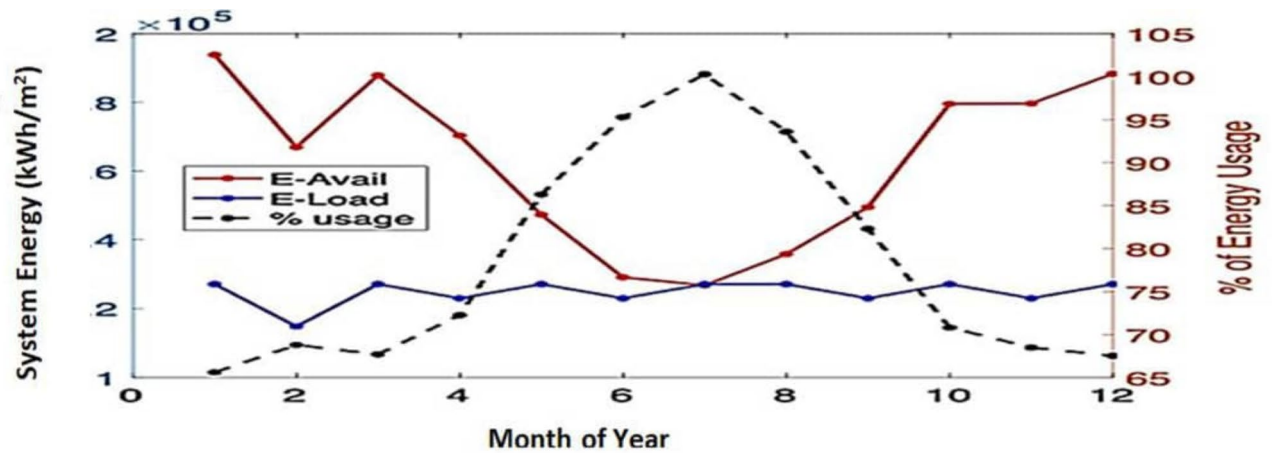


Fig. 9. Energy usage generation profiling of solar and wind resources.

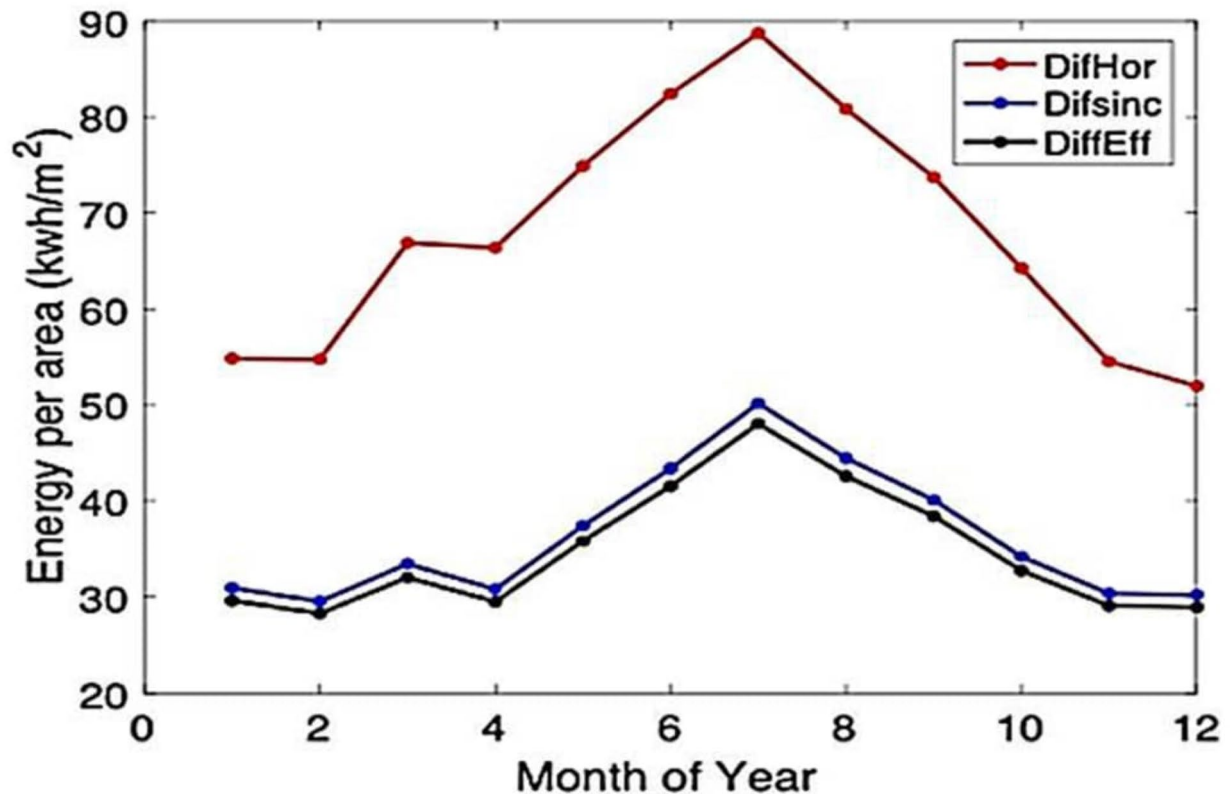


Fig. 10. Diffused irradiation generation profiling of solar and wind resources.

diminish the sensitivity of the solar. The scenario also means that if thermovoltaic or collectors are used, it is better to raise them to a certain height for higher efficiency. The final PV array yield (Ya) and PV system yield (Yr) are presented in Fig. 11. It is observed that the final PV array yield is not dependent on seasonal influences. More so, their maximum is almost half the maximum of the PV system yield. The PV system yield is dependent on seasonal influences with the lowest in July and highest in January. This is due to cloud cover and atmospheric shading which are more in July.

The battery output voltage (U-Batt) and battery efficiency (EffBatt) are presented in Fig. 12. The battery output voltage is sensitive to seasonal influences because of the diffused irradiances presented in Fig. 10. There was a slight abnormality in the minimum battery output voltage in August. The cloud cover was reported to be highest in July, so should the battery output voltage be but there is a slight difference. This abnormality shows that the battery starts diminishing at sharp and prolonged changes in the solar irradiation. This assumption does

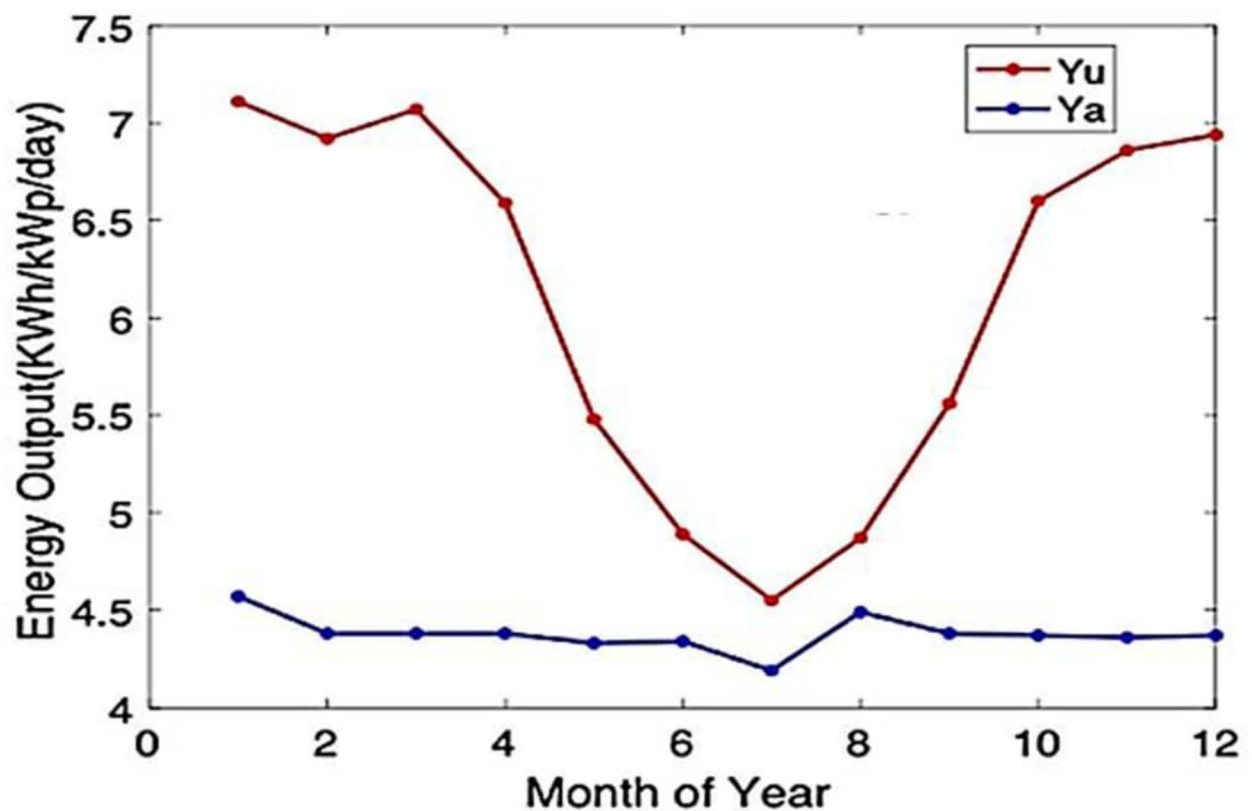


Fig. 11. Final PV yields generation profiling of the proposed hybrid system.

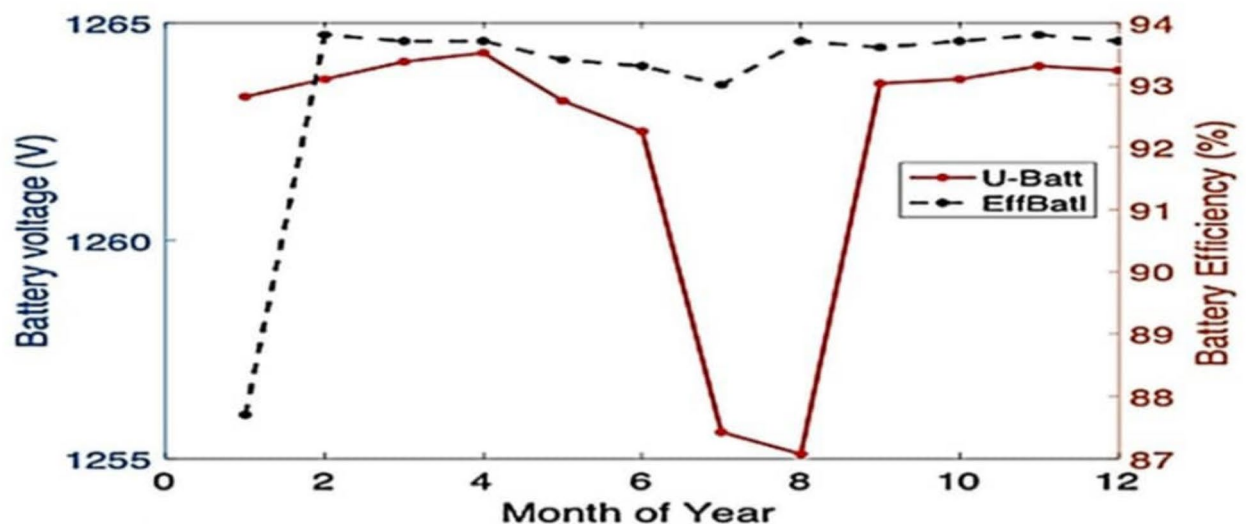


Fig. 12. Battery performance regarding energy storage and dissipation.

not collaborate with the battery efficiency presented in Fig. 12. The variance is linked to the battery type which plays a significant role in storing charges from the charge controller. The currents in the system are discussed in Fig. 13. The array current was seen to be at the maximum compared with other currents in the system. Both currents in the batch mode i.e. battery charging current (I_{Batch}) and battery discharging current (I_{BatD}) are slightly greater than each other at ± 3 Ah. The array current was observed to be five times greater than the current in the batch mode. The implication is the preference to operate a small-scale grid system rather than a standalone system for optimal usage of solar resources in the research area. The battery self-discharge current

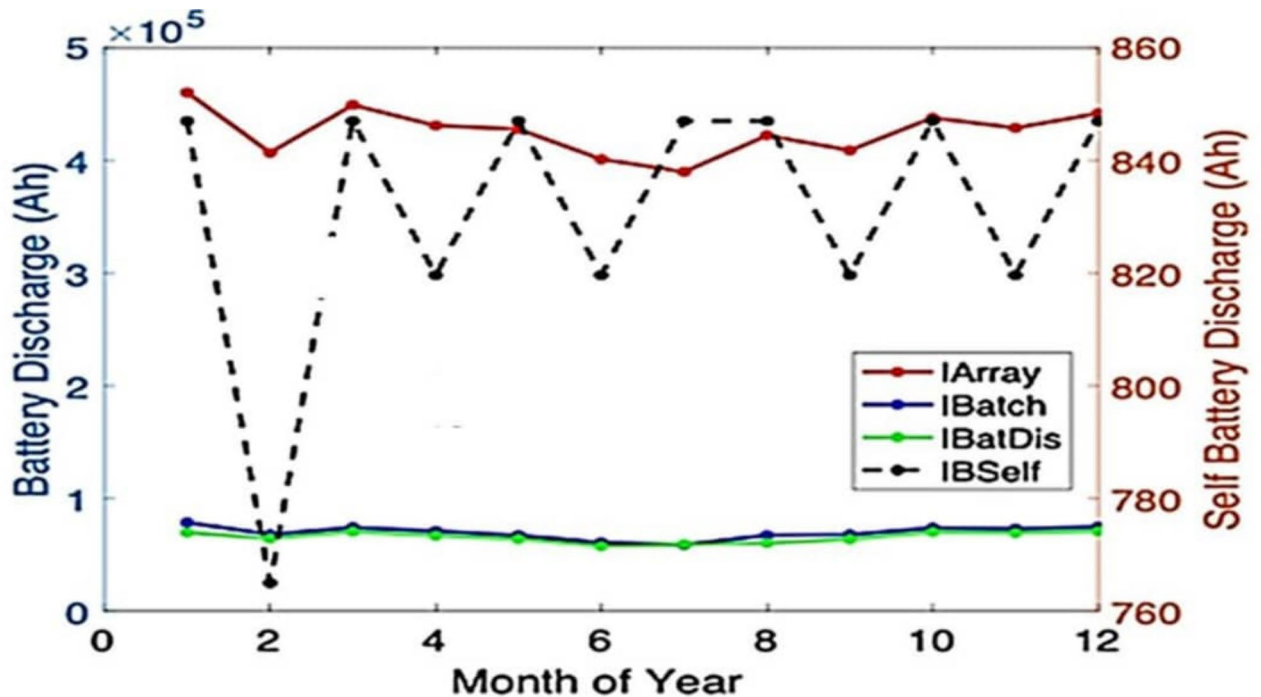


Fig. 13. Current storage and dissipation in the system.

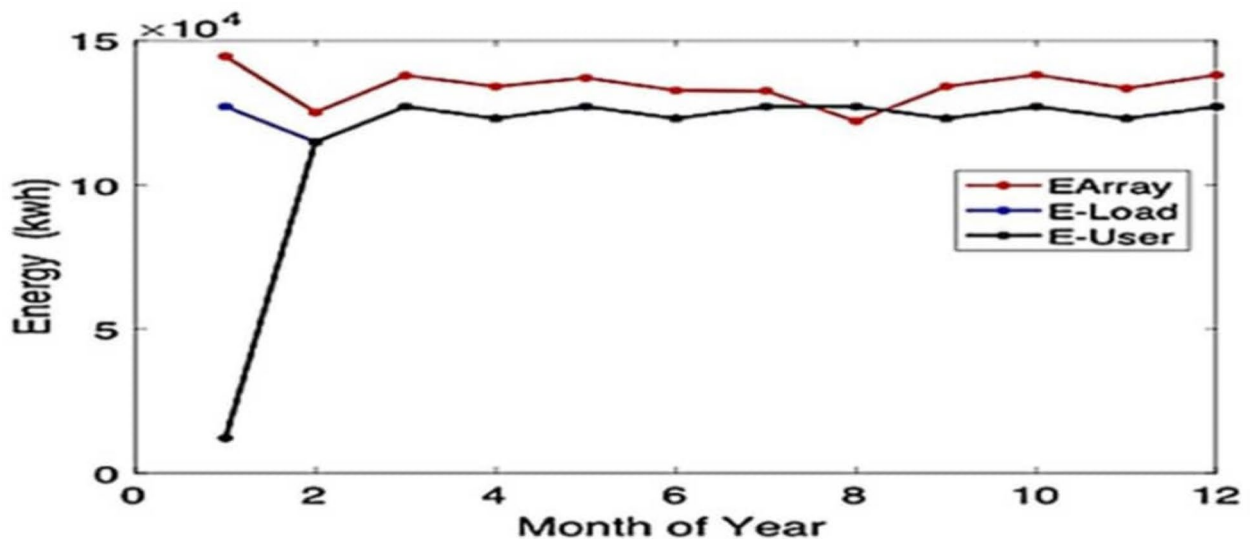


Fig. 14. Energy usage in the system.

(IBSelf) is very low compared to the array current, battery charging current, and battery discharging current, however, the pattern of the self-discharge shows sinusoidal fluctuations that are irregular at the onset and stable when the system is fully functional.

The effective energy at the output of the array (EArray), energy needed by the user (E-Load), and energy supplied to the user (E-User), is displayed in Fig. 14. At the onset of operation, the energy need of the user and energy supplied to the user differs but when the system is fully functional, both energies are the same. There is a disadvantage in this type of scenario; the charging of the battery will be slower as the energy needed by the user and the energy supplied to the user is the same. The slightly higher effective energy at the output of the array reveals that the energy sent to the battery is less than 83.33 Ah for a 12-volt battery and 41.67 Ah for a 24-volt battery. The array virtual energy at a fixed voltage (EArUfix), unused energy (full battery) loss (EUnused), and effective energy at the output of the array (EArray) are presented in Fig. 15. The array virtual energy at a fixed voltage and unused energy (full battery) loss has the same pattern and all depends on the

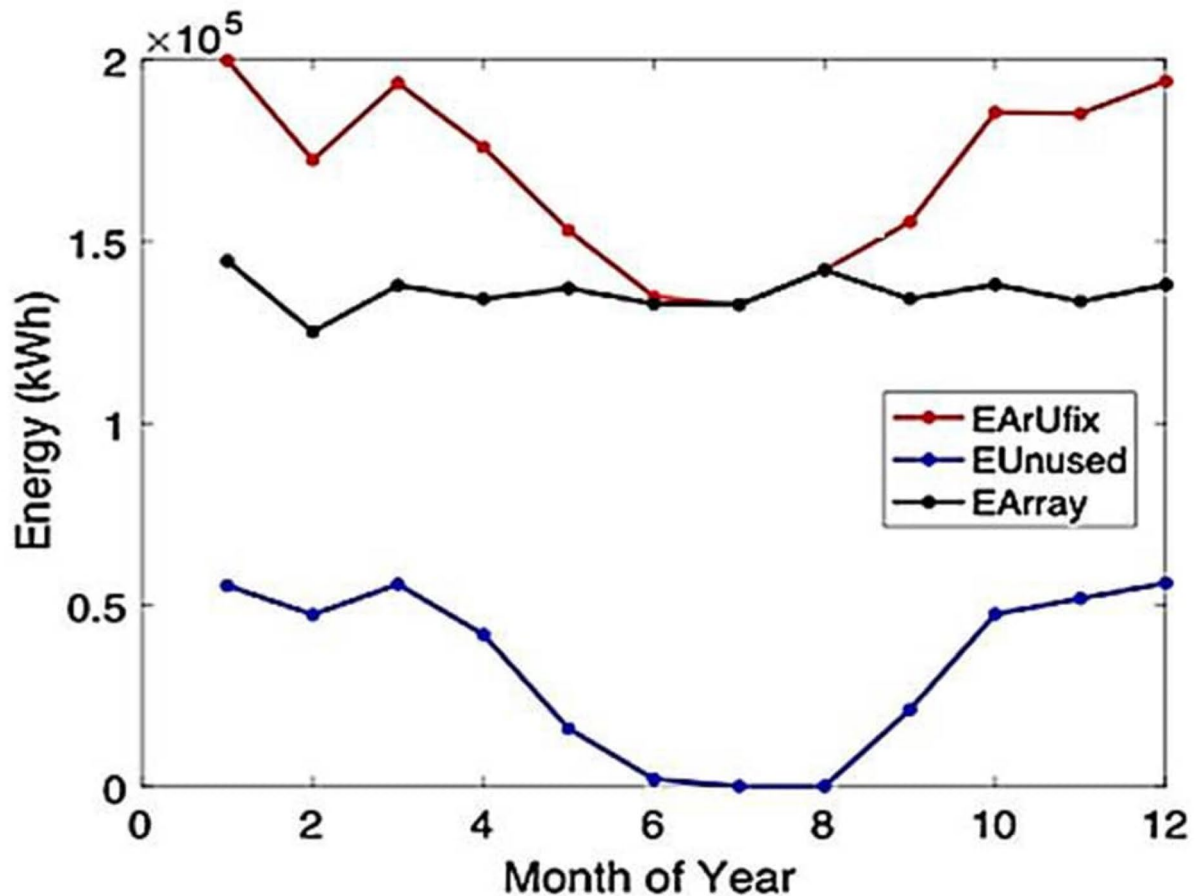


Fig. 15. Energy losses in the system.

seasonal influences as the irradiance. It was observed to be about 25% loss which may be huge considering the set-up cost of a standalone system. The effective energy at the output of the array was more stable but needed more stability with the use of an efficient battery type. The potential of the wind energy in the research location was considered in Fig. 16. The wind speed fluctuates sinusoidally with respect to the seasonal influences. With a maximum wind speed of 4.1 m/s between May and June, an average of 2.7 m/s yearly and a minimum of 1.6 m/s are good indicators to incorporate wind energy into the hybrid system in agreement with the findings of²⁴. The temperature of the location is somewhat high due to current climate change, hence, the need to space hybrid system devices to prevent overheating. The module quality loss (ModQual), module array mismatch loss (MisLoss), and Ohmic wiring loss in the DC array (OhmLoss) are presented in Fig. 17. The module quality loss and Ohmic wiring loss were found to be inversely proportional to each other. The use of quality electrical wiring may reduce the ohmic wiring loss while a well-ventilated room will help reduce the module quality loss. However, the losses from the module quality loss are not expected to reduce drastically because of the well-ventilated room because of the relative air temperature recorded in Fig. 16.

The transposition factor (FTransp), IAM factor on the beam component (FIAMBm), and IAM factor on the global component (FIAMGl) are presented in Fig. 18. The three ratios are seasonally dependent. While the transposition factor is very sensitive to seasonal influences, the IAM factor on the beam component and the IAM factor on the global component are weakly sensitive to seasonal influences. A wide transposition factor shows how the specific patterns of Incident global irradiation in the collector plane (GlobInc) and horizontal global irradiation (GlobHor) could lead to a smooth curve despite the realities of climate change. The IAM factor on the beam component and IAM factor on the global component were almost the same with a slight difference of ± 0.2 because of a slight difference in the seasonal changes in the beam and global component respectively. The average state of charge during the period (SOCmean) and state of charge at the end of the time interval (SOCend) are presented in Fig. 19. Recall that it was reported earlier that the hybrid system has fluctuations at the onset but stabilizes as the operation continues hence, the reason why the state of charge at the beginning of the time interval was not considered in the analysis. However, it could be induced from the differences that the state of charge at the beginning of the time interval was higher than the state of charge at the end of the time interval by about 42%. The normalized unused energy (Lu) and normalized array losses (Lc) are presented in Fig. 20. It is observed that both parameters are influenced by season. More so, the implication of normalized array losses being higher than the normalized unused energy further underscores the huge losses that occur

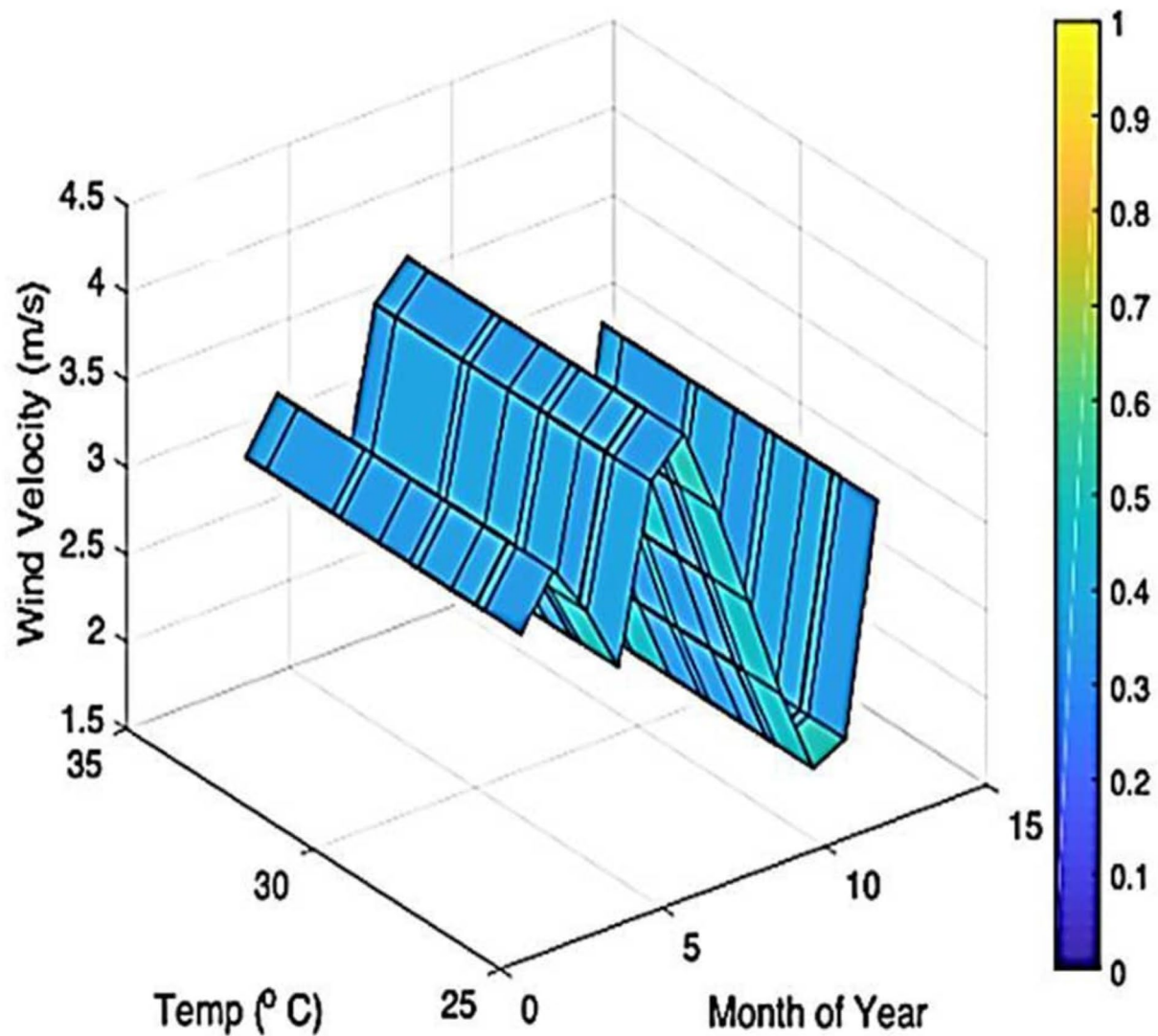


Fig. 16. Wind velocity potential and implication.

in a standalone system. These losses could be controlled in a small smart grid system as storage of unused energy would be one of the priorities to meet users' demands^{25–27}. This assertion is further supported by the normalized system losses in Fig. 21. The normalized system losses are significant i.e., about 600% higher than the normalized unused energy and array losses. The Performance ratio (PR) is dependent on seasonal influences but very effective as the accruable wind energy augments the solar energy drop in July i.e. supported by the horizontal beam irradiation (BeamHor) in Fig. 21.

The financial implication of the continuous operation of the hybrid system for 21 years is presented in Fig. 8. The gain in the hybrid system is presented in Fig. 22. The gain in the long run is expected to be > 400% for all standalone systems. In third-world countries, the standalone system is preferred as the population provides basic amenities for themselves. However, if an energy company should invest in the research location via a small smart grid system, the gain of the project in the next 21 years may be > 600%. Figure 23 reveals that the introduction of high taxes could scuttle the gains of the hybrid standalone system. Hence, for the Government of Gambia to encourage investment in renewable energy users, the taxes on the purchase of items should be deliberately low so that the purchasing and maintenance costs can be affordable. Also, it is expected that users should have access to loans at low interest to be able to invest in clean energy hybrid systems (Fig. 24). Due to the losses mentioned above, the running cost is expected to increase linearly throughout the 21 years due to periodic maintenance over time (Fig. 25). When big energy investors come into the research location, it is expected that the electricity sales to users will be constant while the taxable income will gradually diminish due to the maintenance cost of the facility.

In Table 3, the present study results have been compared with other recent research papers done using similar simulation tools in other locations. And our simulation results show that the optimal size was possible with only 20 photovoltaic modules (250 W) and 1 Wind generator (1 KW) compatible with the annual load requirement

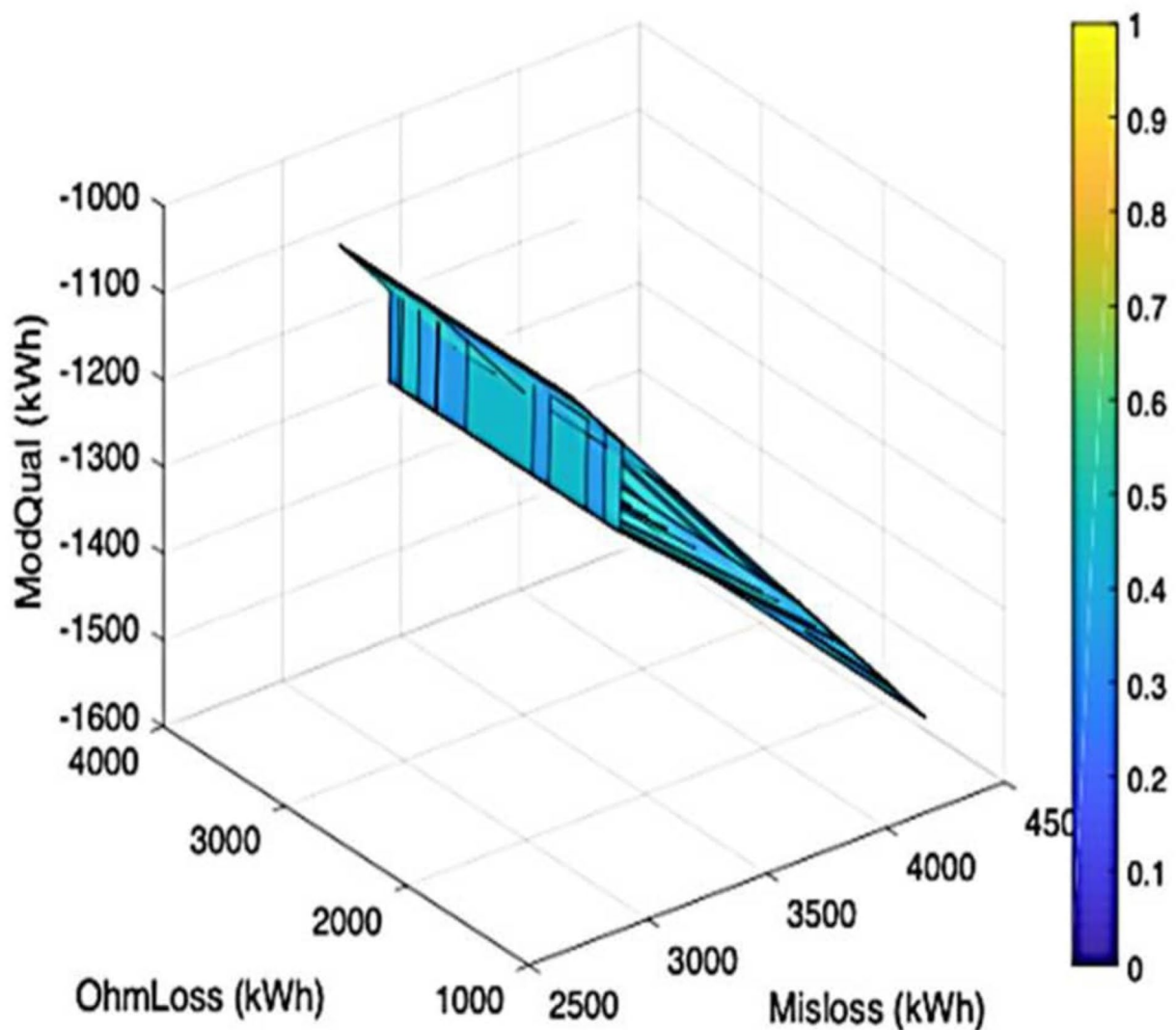


Fig. 17. System losses and implication.

of about 2555 MWh which is cost-saving relative to similar systems that requires more photovoltaic panels and other renewable energy sources to achieve similar or even less annual loads requirement^{14, 16, 17, 23}. Furthermore, the robust inclusion of the real-time cost of installation and electricity sale in the Gambia has projected that the operation of the hybrid system for 21 years presents a net gain of > 400% for the standalone system making it an ideal choice for investors in the power sector.

Conclusion

Technical and economic simulation of a proposed hybrid renewable energy power system has been carried out in the Basse Santa Su district area of The Gambia. It was shown that the effective energy from solar increases by the day with an average of 7 MW effective energy achievable per day or (2555 MWh) annually which implies that the solar energy of the location can be conveniently scaled up for industrial and commercial application.

Our findings revealed that the solar fraction remains constant throughout the year while the performance ratio was seen to be dependent on seasonal variables. The proposed hybrid system performance has the highest values in July with a build-up from April as the effective energy was seen to increase with respect to the effective power. It was observed that the variable array power distribution increased sinusoidally to about 900 kW before dropping rapidly. This implied that the system setting diminishes due to significant ohmic and thermal losses. In addition, at the onset of operating the system, it was observed that the energy need of the user and energy supplied to the user differs but when the system is fully functional, both energies are the same. This can be a disadvantage because, under such conditions, the charging of the battery will likely be slower as the energy need of the user and energy supplied to the user is the same which is a shortcoming of the proposed system. The effective energy at the output of the array was more stable but yet we suggested that the stability may increase

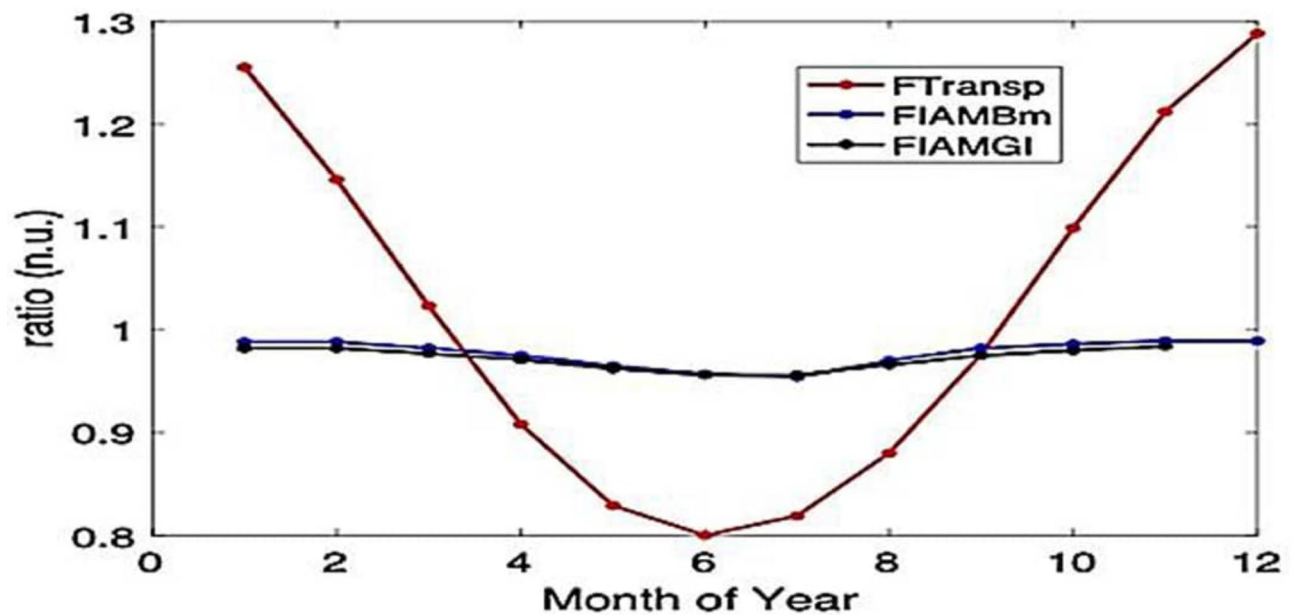


Fig. 18. Global and beam radiation influences on system ratio operation.

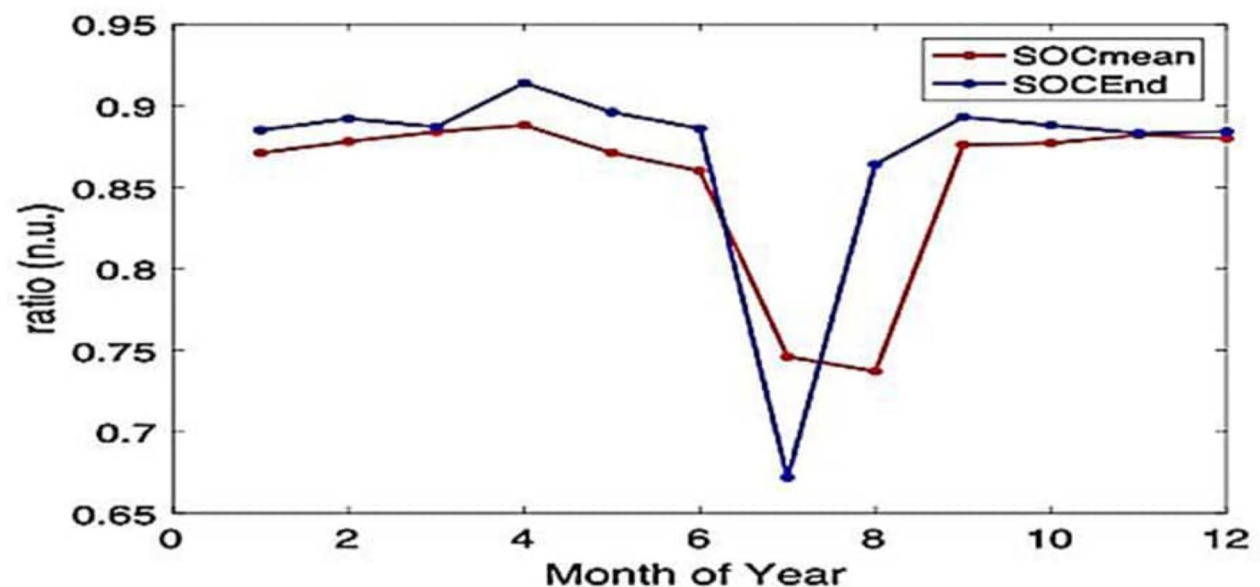


Fig. 19. Time analysis of the system ratio operations.

with the use of a more efficient battery type. Furthermore, it was observed that the wind speed fluctuates sinusoidally with respect to the seasonal and meteorological influences; with a maximum wind speed of 4.1 m/s between May and June, an average of 2.7 m/s yearly, and a minimum of 1.6 m/s which are good indicators to incorporate into the hybrid system. Further analysis on incorporating wind into the proposed HREPS revealed that the most probable power output from the wind in the study location was recorded at a wind speed of about 3–4 m/s with peak power virtually anticipated at the wind speed of 7 m/s which was never observed in reality. The temperature of the location is somewhat high due to current climate change, hence, there is a need to space the hybrid system devices to prevent overheating.

Economically, the financial implication of the continuous operation of the hybrid system for 21 years presents a net gain of greater than 400% for the standalone systems and this result is in line with the recommendations of²⁶. Since in third world countries, most of the inhabitants provide basic amenities like electricity for themselves,

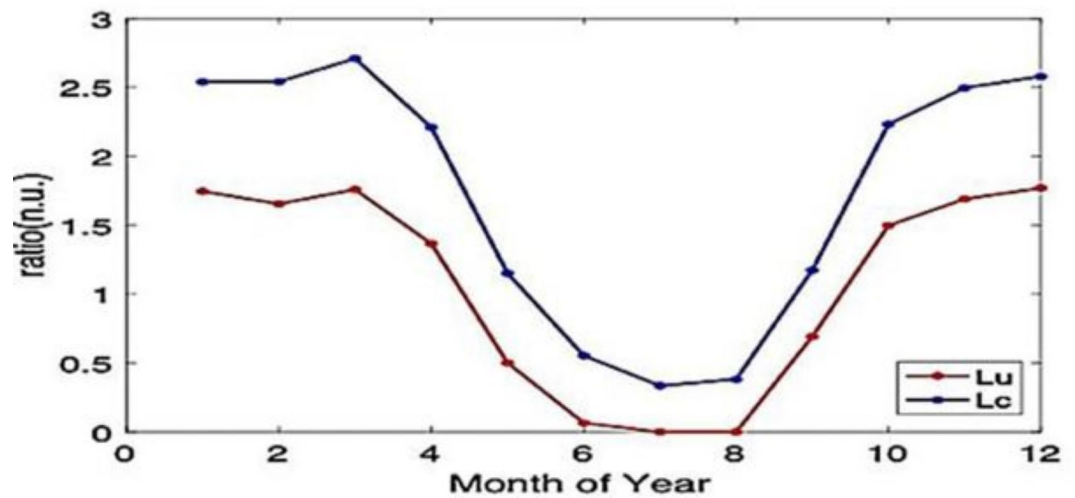


Fig. 20. External losses from system ratio operations.

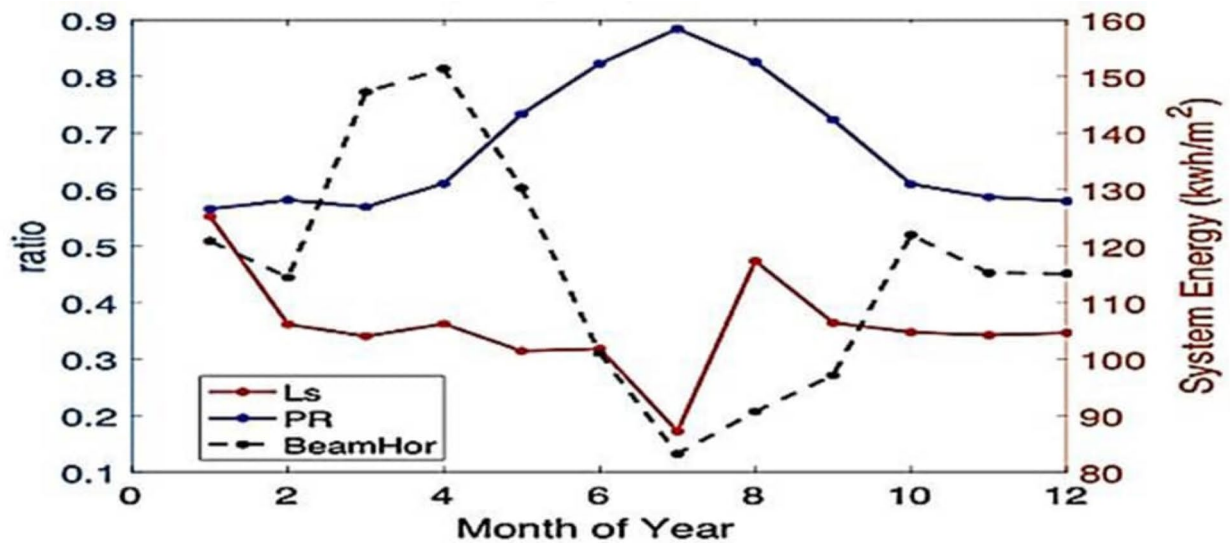


Fig. 21. Effectiveness of hybrid system ratio operations.

this type of standalone system may be considered a first choice option to explore. However, if an energy company should invest in the research location via a small smart grid system, the gain of the project in the next 21 years may be > 600%. The simulation results of the electricity sales to users were seen to be constant with the taxable income gradually diminishing due to the maintenance cost of the facility. The economic analysis further revealed that the introduction of high taxes could scuttle the gains of the hybrid standalone proposed system. Hence, for the Government of The Gambia to encourage investment in renewable energy power systems, the taxes on the purchase of items should be deliberately low so that the purchasing and maintenance costs can be affordable.

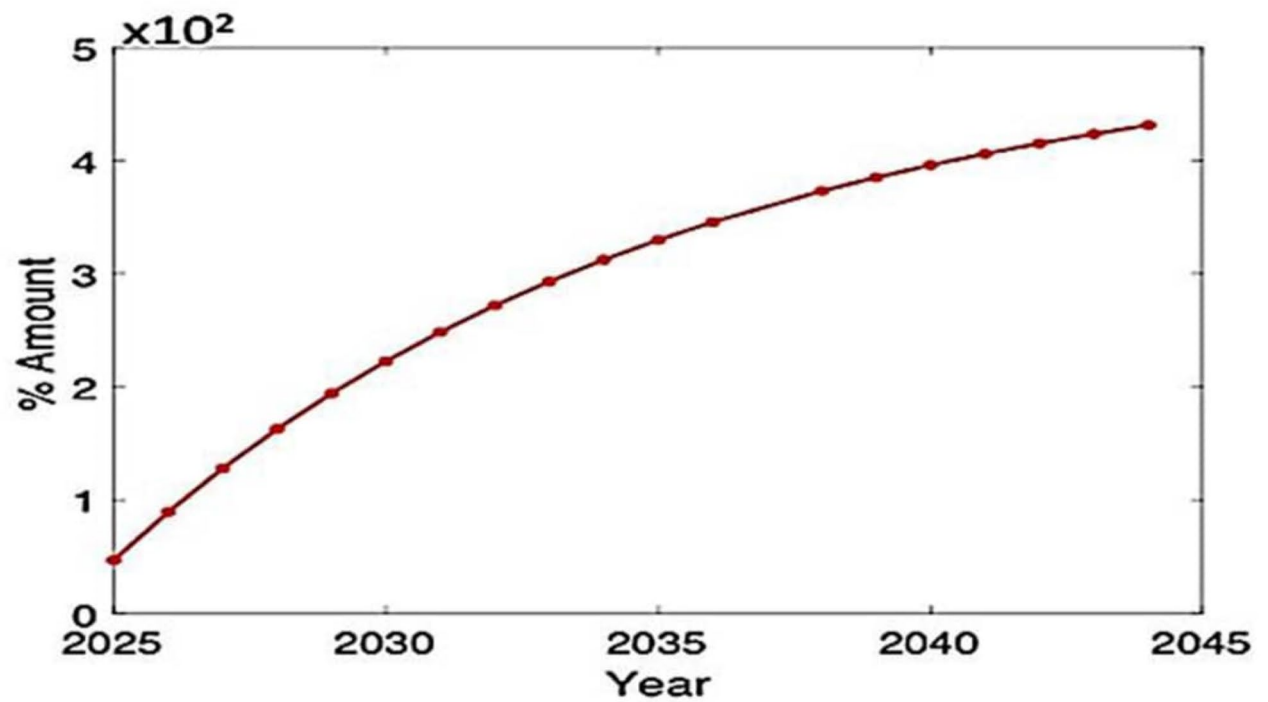


Fig. 22. Hybrid system gain forecast and implication.

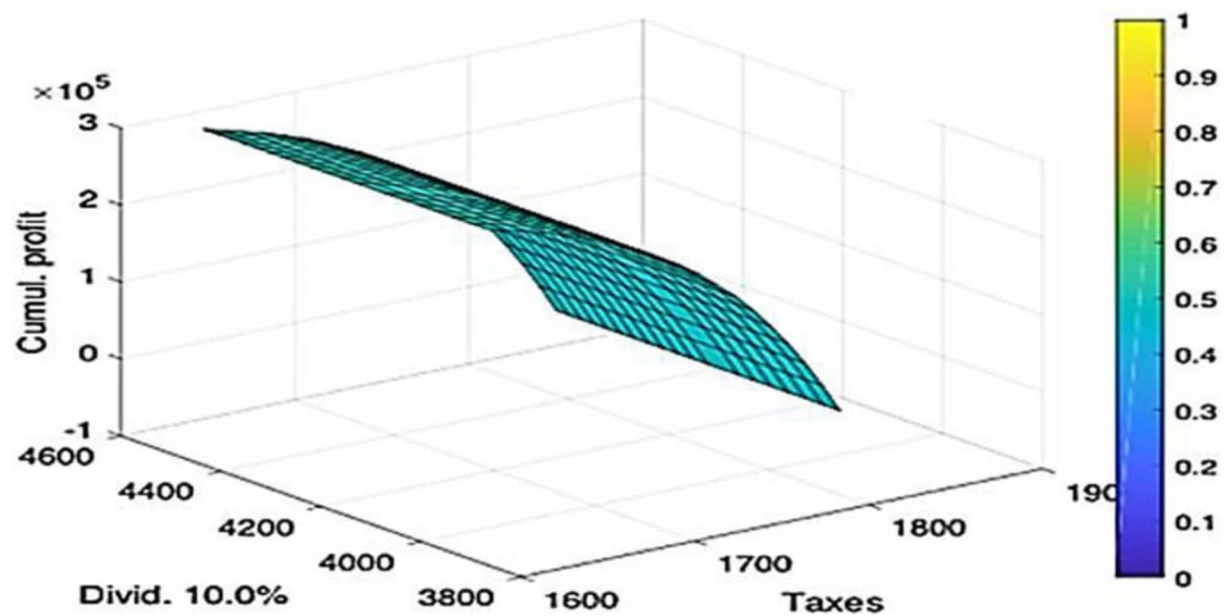


Fig. 23. Forecast of the implication of taxes on the cumulative gain of the proposed system.

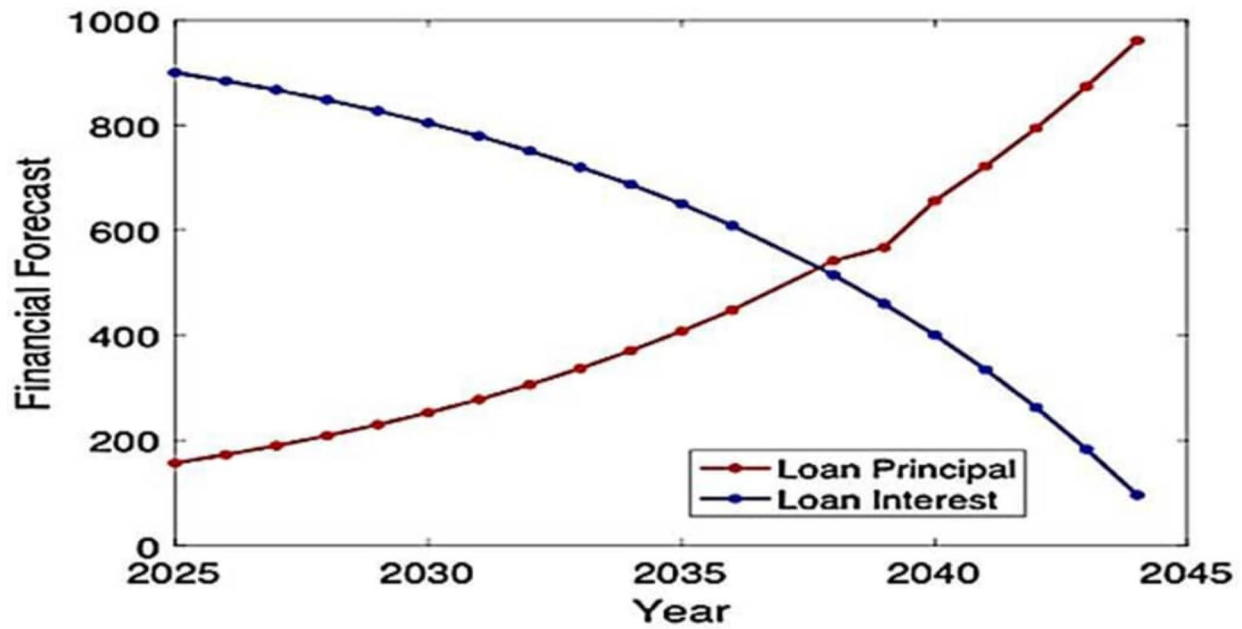


Fig. 24. Forecast of the implication of interest rate on loans to invest in the proposed system.

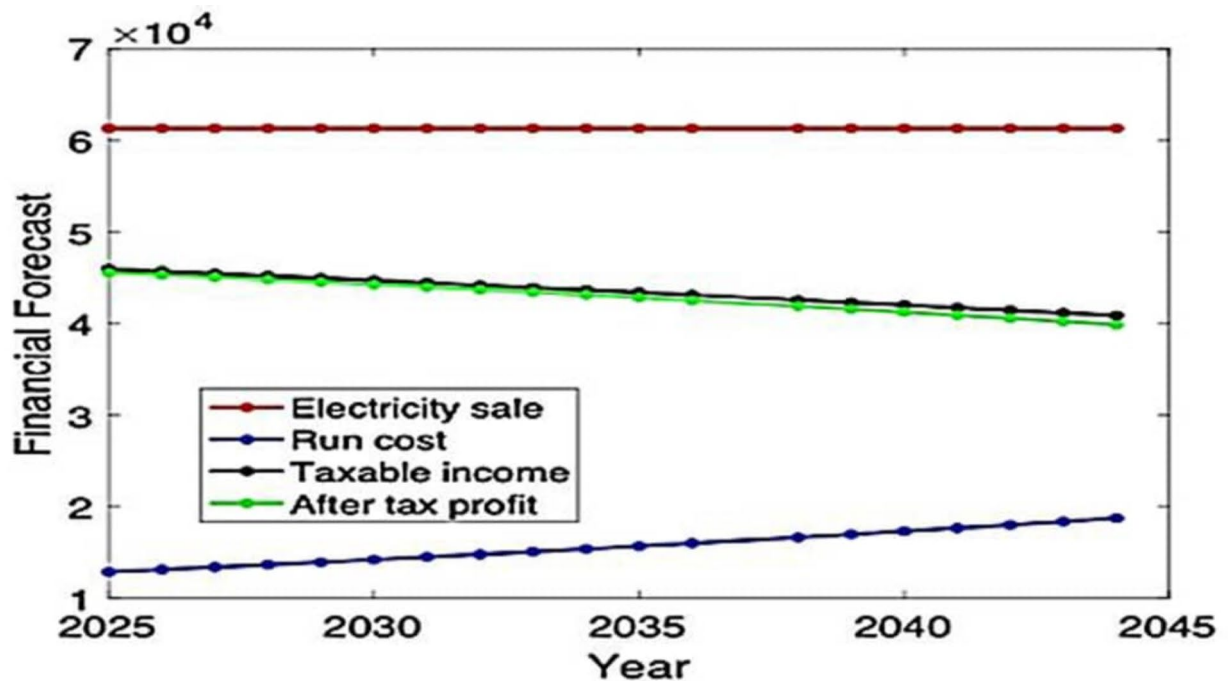


Fig. 25. Financial forecast of running cost losses and implications on the expected profit.

S/N	Author(s)	Research	Methodology	Location
1	Tarigan et al. ²³	Techno-Economic Simulation of a Grid-Connected PV System Design as Specifically Applied to Residential Houses	The simulation in this study was carried out for a 1 kWp capacity of grid-connected PV system using simulated with RETScreen and PVsyst	Surabaya, Indonesia
2	Mulumba and Farzaneh ¹⁴	A dynamic techno-economic simulation and analysis of a hybrid renewable energy power (solar-wind-battery-flywheel) system in remote areas	Multiple objective optimization modeling techniques and the epsilon (€) approach. And use of Matlab/Simulink simulation	Remote areas in Kenya
3	Eze et al. ¹⁷	Technical and economic feasibility assessment of hybrid renewable energy system (Grid/wind/photovoltaic/diesel/battery/converter) of an institutional building	Application of HOMER™ Pro microgrid analysis tool using local meteorological data.	Nairobi, Kenya
4	Samy ¹⁶	Techno-Economic Analysis of Hybrid Renewable Energy Systems (solar PV/Wind/Fuel Cell) for Electrification	Mathematical modeling, simulation, and optimization approach	Beni-Suef province, Egypt.
5	Doile et al. ²⁷	Assess the economic feasibility of distributed hybrid power plants with battery banks	Stochastic analyses using many hybrid power plant variables of up to 1 MW and small power plant rating of 5 MW installed power).	Brasilia, Brazil
6	Ayua and Emeter [This study]	Technical and Economic Simulation of a Hybrid Renewable Energy Power System Design for Industrial Application	The PVsyst and Matlab code was used to evaluate the simulation results of the proposed hybrid renewable energy system to identify the best technical and economical resolution taking into account the meteorological variables of the location	Basse Santa Su, The Gambia

Table 3. Comparison of the present study with other recent research papers done using similar tools.

Data availability

The data used in this study was generated and will be made available by the corresponding author upon reasonable request.

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Author contributions

Author contribution statement T. J. Ayua contributed to the study conception design, methodology, wrote the original draft and reviewed the manuscript. M. E. Emeter performed the data analysis, wrote the paper, and reviewed the manuscript. All the authors have read and agreed to the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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